



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 09: Population Principal Component Analysis

Lecture 01: Introduction to Principal Component Analysis



High-Dimensional Data Challenges

Problem

Many datasets have thousands or millions of features, making computation and interpretation difficult.

Curse of Dimensionality

As dimensions increase, points become sparse and distances less informative, complicating analysis.

Impact of High Dimensionality

Overfitting

High-dimensional data can cause models to capture noise rather than patterns.

Poor Generalization

Models trained on sparse data often perform poorly on new, unseen data.

Visualization Difficulty

High-dimensional datasets are hard to visualize and interpret directly.

Dimensionality Reduction Overview

Definition

Dimensionality reduction reduces the number of features while retaining important information.

Benefits

- Simplifies datasets
- Improves computational efficiency
- Reduces overfitting
- Enables visualization in 2D or 3D

Principal Component Analysis (PCA)

What is PCA?

PCA is a technique for reducing dimensions by projecting data onto directions of maximum variance.

Benefits of PCA

Variance Preservation

Captures directions of maximum variance while discarding noise.

Efficiency and Overfitting

Reduces computational cost and prevents overfitting by removing redundant features.

Visualization

Enables projecting high-dimensional data into 2D or 3D for easier interpretation.

Data Centering in Population PCA

Why Center Data?

- Principal Component Analysis (PCA) is a geometric procedure that identifies the directions of maximum variance in the data.
- Variance is always measured relative to an origin. In population PCA, we need the origin to represent the "true center" of the data distribution.
- If the data are not centered, the covariance matrix includes cross-terms from the population mean, distorting the geometric interpretation of the spread.
- To define a coordinate system intrinsic to the population, we shift the origin to the population mean.
- Suppose $X_1, X_2, \dots, X_n, \dots$ are p -dimensional random vectors drawn from the population. The population mean vector is defined as

$$\mu = \mathbb{E}[X] \in \mathbb{R}^p,$$

which represents the expected value of a typical observation X from the population.

- Center the data by subtracting the population mean:

$$\tilde{X} = X - \mu,$$

so that $\mathbb{E}[\tilde{X}] = 0$. The origin now lies at the center of the population distribution.

Covariance Matrix

Measuring Relationships Between Features

- After centering the data, we want to understand how the features vary together.
- The covariance between two features measures whether they increase or decrease together.
- Formally, the covariance matrix is defined as

$$\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X} \in \mathbb{R}^{p \times p}.$$

Interpreting Covariance Entries

- Diagonal entries Σ_{jj} give the variance of feature j .
- Off-diagonal entries Σ_{jk} give the covariance between features j and k .
- Large positive covariance means two features increase together; large negative covariance means one increases while the other decreases.

Bivariate Normal Distribution before Centering

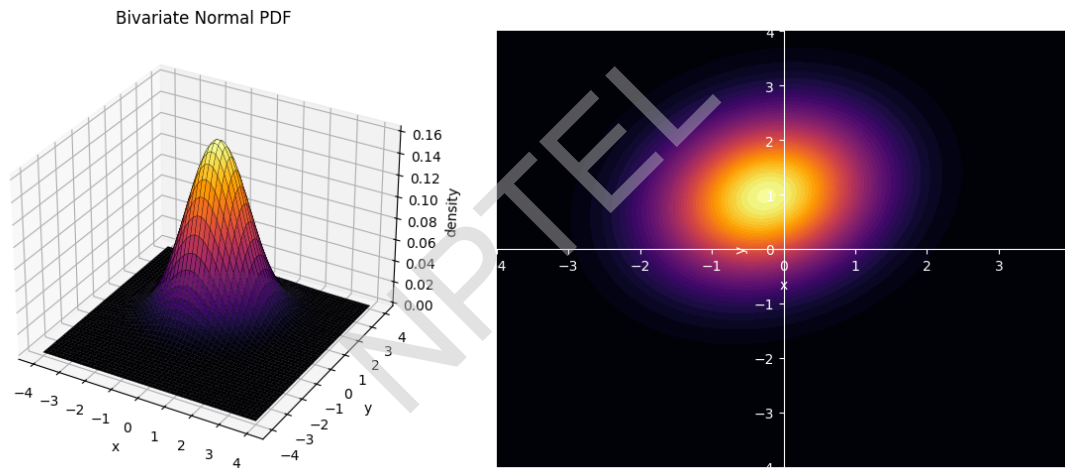


Figure: Bivariate Normal Distribution with $\mu_X = -0.24$ and $\mu_Y = 0.975$, $\sigma_X = 1$ and $\sigma_Y = 1$, $\rho = 0.162$.

Bivariate Normal Distribution after Centering

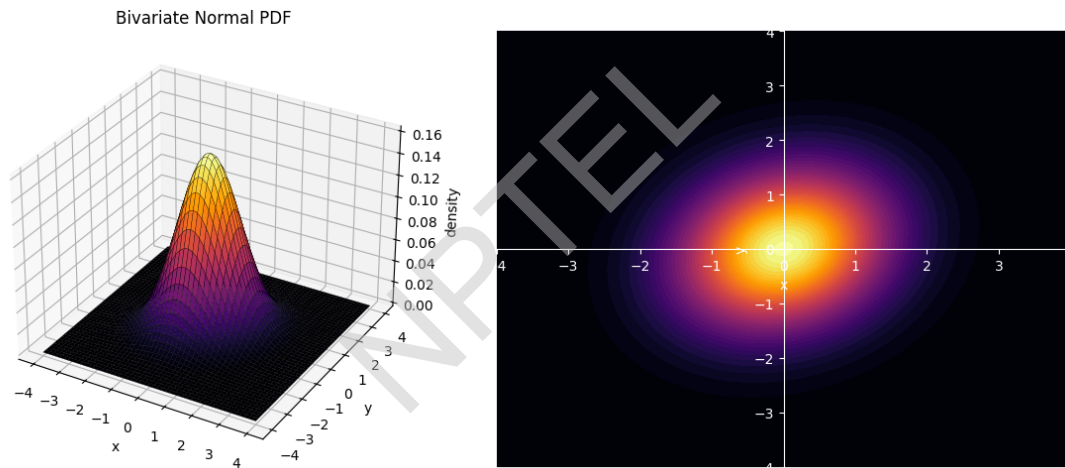


Figure: Bivariate Normal Distribution with $\mu_X = 0$ and $\mu_Y = 0$, $\sigma_X = 1$ and $\sigma_Y = 1$, $\rho = 0.162$.

Bivariate Normal Distribution

Definition of Bivariate Normal Distribution

A random vector $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in \mathbb{R}^2$ is said to have a bivariate normal distribution if its probability density function is:

$$f(\mathbf{X}) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{X} - \mu)^\top \Sigma^{-1}(\mathbf{X} - \mu)\right)$$

Where:

- $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$ is the *mean vector*,
- Σ is the *covariance matrix*.

Centered Data Notation

Center the data by subtracting the mean from each row:

$$\tilde{X} = X - \mathbf{1}_n \mu^\top$$

After centering, each column of $\tilde{X}$ has zero mean, so the origin now lies at the center of the data cloud. PCA axes (eigenvectors of the covariance) are then computed in this intrinsic coordinate system.

Contours of the Bivariate Normal Distribution

Contours of Constant Density

In the centered coordinate system $\tilde{X}$, the contours of constant density are ellipses given by:

$$\tilde{X}^\top \Sigma^{-1} \tilde{X} = 2c$$

These ellipses are now centered at the origin of the centered data cloud.

Geometric Interpretation

The *major axis* corresponds to the direction of maximum variance, and the *minor axis* corresponds to the direction of minimum variance. The shape and orientation of these ellipses are entirely determined by the covariance matrix Σ .

Eigenvalue Decomposition of the Covariance Matrix

Eigenvalue Decomposition

Diagonalizing the covariance matrix Σ in the centered space gives:

$$\Sigma = Q\Lambda Q^T$$

Where:

- Q is the matrix of *eigenvectors* (principal components),
- $\Lambda = \text{diag}(\lambda_1, \lambda_2)$ is the diagonal matrix of *eigenvalues*.

Principal Components in Centered Space

Principal Components and Variance

- The eigenvectors of Σ define the *directions* of the PCA axes in the centered coordinate system $\tilde{X}$.
- Each eigenvector corresponds to a direction of *maximum variance* in the data, after projecting out higher-variance directions.
- The corresponding eigenvalue quantifies the *spread of the data* along that direction.
- The first principal component (eigenvector with largest eigenvalue) points along the *direction of greatest variance*.
- Subsequent principal components point along orthogonal directions of decreasing variance.

Effect of Correlation on the Covariance Matrix

Covariance Matrix with Correlation

For centered data $\tilde{X}$ with correlation coefficient ρ between X_1 and X_2 :

$$\Sigma = \begin{pmatrix} \sigma_{11} & \rho\sigma_{11}\sigma_{22} \\ \rho\sigma_{11}\sigma_{22} & \sigma_{22} \end{pmatrix}$$

Where:

- $\sigma_{11} = \text{Var}(X_1)$,
- $\sigma_{22} = \text{Var}(X_2)$,
- $\rho = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)\text{Var}(X_2)}}$ is the *correlation coefficient*.

Effect of Correlation

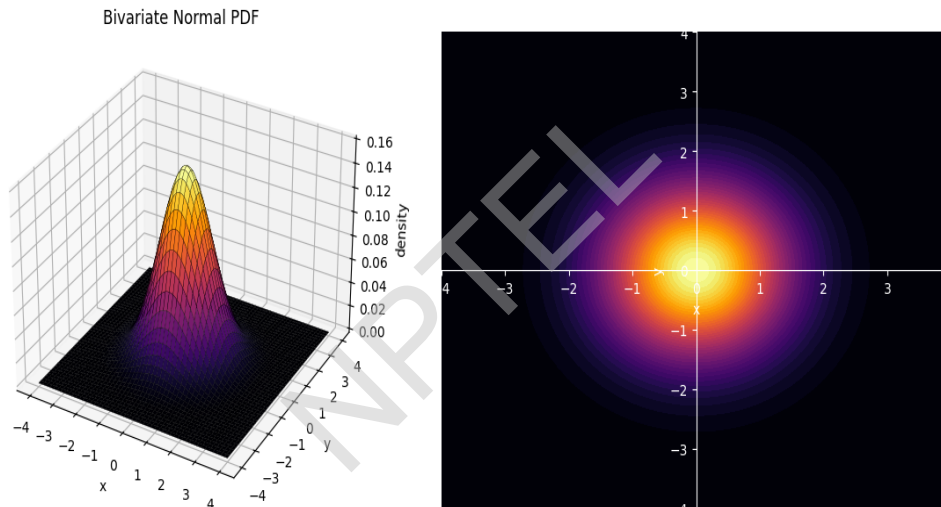


Figure: Bivariate Normal Distribution with $\mu_X = 0$ and $\mu_Y = 0$, $\sigma_X = 1$ and $\sigma_Y = 1$, $\rho = 0$.

Note:

$\rho = 0$: *axis-aligned ellipses* (circular if variances are equal),

Effect of Correlation

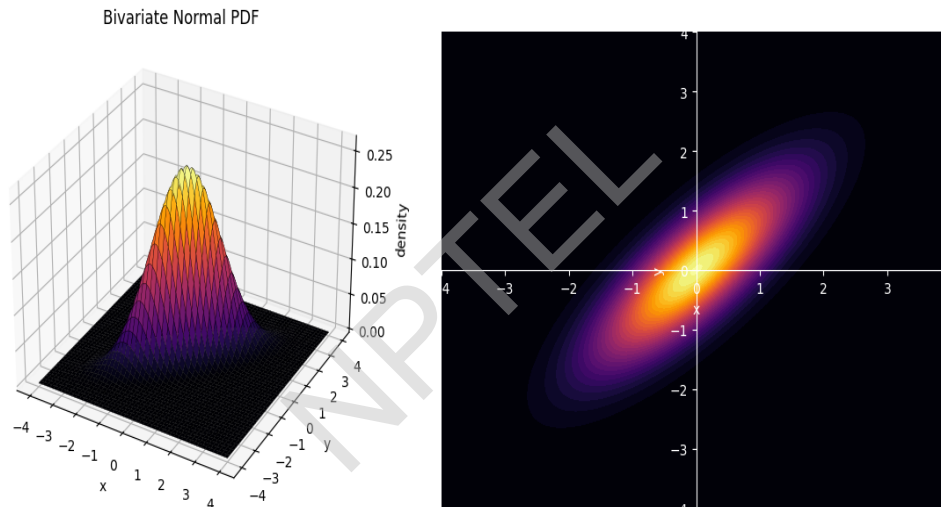


Figure: Bivariate Normal Distribution with $\mu_X = 0$ and $\mu_Y = 0$, $\sigma_X = 1$ and $\sigma_Y = 1$, $\rho > 0$.

Note:

$\rho > 0$: ellipses tilted along the upper-right diagonal.

Effect of Correlation

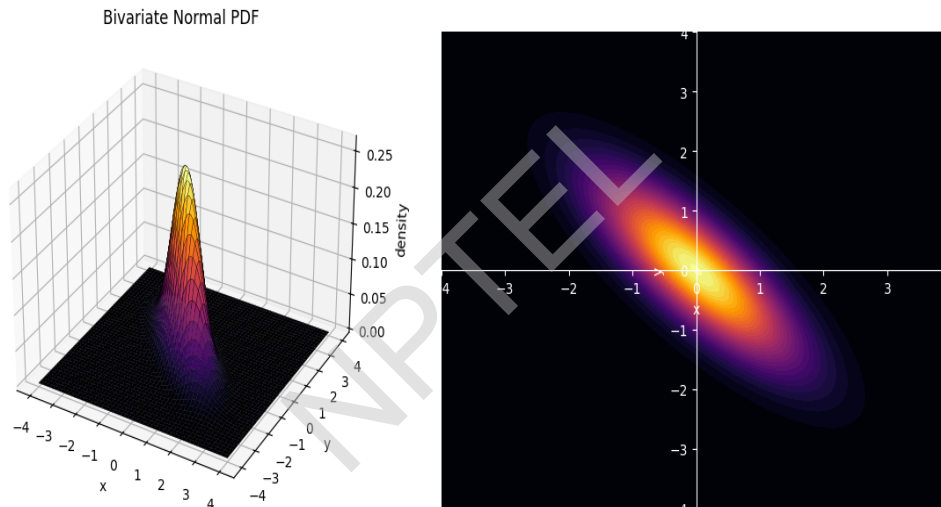


Figure: Bivariate Normal Distribution with $\mu_X = 0$ and $\mu_Y = 0$, $\sigma_X = 1$ and $\sigma_Y = 1$, $\rho < 0$.

Note:

$\rho < 0$: ellipses tilted along the upper-left to lower-right diagonal.

Back to Original Coordinates

Ellipse Axes in X -space

The ellipse in the original (X_1, X_2) -space is described by:

$$\text{Major axis direction} = \mathbf{q}_1, \quad \text{Length} = \sqrt{c \lambda_1}$$

$$\text{Minor axis direction} = \mathbf{q}_2, \quad \text{Length} = \sqrt{c \lambda_2}$$

Where:

- $\mathbf{q}_1$ and $\mathbf{q}_2$ are *eigenvectors* of Σ ,
- λ_1 and λ_2 are *eigenvalues*, representing the *spread* along each principal axis.

Principal Component Analysis (PCA) in Centered Space

Goal of PCA

PCA rotates the centered data $\tilde{X}$ to new *orthogonal axes* (principal components) where the covariance matrix is diagonal, making the transformed variables uncorrelated.

Steps in PCA Using $\tilde{X}$

- 1 Center the data: $\tilde{X} = X - \mathbf{1}_n \mu^\top$
- 2 Compute the covariance matrix: $\Sigma = \frac{1}{n} \tilde{X}^\top \tilde{X}$
- 3 Perform eigenvalue decomposition: $\Sigma = Q \Lambda Q^\top$
- 4 Transform data to principal components: $Y = Q^\top \tilde{X}$

Interpretation

In this centered coordinate system, the axes align with directions of maximum variance, and the eigenvalues indicate the magnitude of variance along each principal component.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ Python packages (Matplotlib, NumPy, SciPy) are used to generate the bi-variate normal distribution visualizations.

Thank you





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Module 09: Population Principal Component Analysis

Lecture 02: Eigenvalues and Eigenvectors



Matrix as a Function

Matrix as a Vector-to-Vector Function

A matrix $A \in \mathbb{R}^{m \times n}$ can be thought of as a function that takes a vector in $\mathbb{R}^n$ and produces another vector in $\mathbb{R}^m$:

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

which takes an input vector $x \in \mathbb{R}^n$ and outputs a vector $y \in \mathbb{R}^m$:

$$x \xrightarrow{A} y = Ax$$

Input and Output

- Input: a vector $x \in \mathbb{R}^n$
- Output: a vector $y = Ax \in \mathbb{R}^m$

So, instead of viewing a matrix as just a rectangular array of numbers, think of it as a **rule for transforming vectors**.

Eigenvectors: Intuitive View

- Think of a matrix as a rule that transforms vectors in space.
- Most vectors change both direction (in the direction of the line joining the origin and the point) and length under this transformation.
- Eigenvectors are special vectors whose direction (in the direction of the line joining the origin and the point) remains the same when the matrix is applied.
- Their length may change, but the arrow from the origin points in the same direction after the transformation.
- They represent the “fixed directions” of the transformation.

Eigenvalues and Eigenvectors: Formal Definition

Eigenvalue and Eigenvector

Let $A_{n \times n}$ be an $n \times n$ square matrix with real entries. A non-zero vector $\mathbf{v} \in \mathbb{R}^n$ is called an *eigenvector* of A if there exists a scalar $\lambda \in \mathbb{R}$, called the *eigenvalue* associated with $\mathbf{v}$, such that:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Eigenvalues and Eigenvectors

- A is a square matrix with dimensions $A_{n \times n}$.
- $\mathbf{v}$ is a non-zero vector in $\mathbb{R}^n$.
- λ is a scalar corresponding to $\mathbf{v}$.

Properties of Eigenvectors

Scalar Multiplication Property

If $\mathbf{v}$ is an eigenvector of the matrix $A_{n \times n}$ corresponding to eigenvalue λ and $c \neq 0$ is a scalar, then $c\mathbf{v}$ is also an eigenvector corresponding to λ :

$$A(c\mathbf{v}) = c(A\mathbf{v}) = c(\lambda\mathbf{v}) = \lambda(c\mathbf{v}).$$

Proof:

Let A be a square matrix and $\mathbf{v}$ be an eigenvector corresponding to eigenvalue λ , i.e.,

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Now, consider a scalar $c \neq 0$. We want to show that $c\mathbf{v}$ is also an eigenvector corresponding to the same eigenvalue λ . Compute $A(c\mathbf{v})$:

$$A(c\mathbf{v}) = cA\mathbf{v} \quad (\text{by the distributive property of scalar multiplication}).$$

Substituting $A\mathbf{v} = \lambda\mathbf{v}$:

$$A(c\mathbf{v}) = c(\lambda\mathbf{v}) = \lambda(c\mathbf{v}).$$

Hence, $c\mathbf{v}$ is an eigenvector of A with eigenvalue λ .

Properties of Eigenvectors

Sum of Eigenvectors

Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors of a matrix $A_{n \times n}$ corresponding to the same eigenvalue λ such that $\mathbf{v}_1 \neq -\mathbf{v}_2$. Then their sum $\mathbf{v}_1 + \mathbf{v}_2$ is also an eigenvector:

$$A(\mathbf{v}_1 + \mathbf{v}_2) = A\mathbf{v}_1 + A\mathbf{v}_2 = \lambda\mathbf{v}_1 + \lambda\mathbf{v}_2 = \lambda(\mathbf{v}_1 + \mathbf{v}_2).$$

Proof:

$$\begin{aligned} A(\mathbf{v}_1 + \mathbf{v}_2) &= A\mathbf{v}_1 + A\mathbf{v}_2 \quad (\text{linearity of matrix multiplication}) \\ &= \lambda\mathbf{v}_1 + \lambda\mathbf{v}_2 \quad (\text{using eigenvector definition}) \\ &= \lambda(\mathbf{v}_1 + \mathbf{v}_2) \end{aligned}$$

Hence, $\mathbf{v}_1 + \mathbf{v}_2$ is an eigenvector corresponding to λ .

Note:

If eigenvectors correspond to different eigenvalues, their sum is generally not an eigenvector.

Eigen Space of an Eigen Value

Definition

Let $A_{n \times n}$ be an $n \times n$ matrix and λ be a fixed eigen value of A . Then the eigen space corresponding to the eigenvalue λ is given by:

$$E_\lambda = \{ \mathbf{v} : A\mathbf{v} = \lambda\mathbf{v} \}.$$

Note:

- For any eigen value λ of an $n \times n$ matrix A , the set E_λ contains the zero vector.
- For a fixed eigen value of an $n \times n$ matrix A , the set E_λ is closed under addition and scalar multiplication.
- For a fixed eigen value of an $n \times n$ matrix A , the set $E_\lambda = \{\text{zero vector}\} \cup \{\text{eigen vectors of } A \text{ corresponding to } \lambda\}.$

Characteristic Polynomial and Equation

Characteristic Polynomial

The *characteristic polynomial* of an $n \times n$ matrix A is defined as the determinant of the matrix $A - tI$, where I is the identity matrix and t is a scalar (indeterminate):

$$p_A(t) = \det(A - tI)$$

This polynomial is a degree n polynomial in the indeterminate t .

Characteristic Equation

The *characteristic equation* is obtained by setting the characteristic polynomial equal to zero:

$$p_A(t) = \det(A - tI) = 0$$

- This equation is used to find the eigenvalues of the matrix A .

Eigenvalues and the Characteristic Equation

Theorem

A scalar λ is an eigenvalue of a square matrix $A_{n \times n}$ if and only if it is a root of the characteristic equation:

$$\det(A - tI) = 0,$$

where I is the $n \times n$ identity matrix, and t is the indeterminate used in the characteristic equation.

Proof:

Step 1: Start with the definition of eigenvalues and eigenvectors.

- Let A be a square matrix of size $n \times n$, and suppose $\mathbf{v}$ is a non-zero vector in $\mathbb{R}^n$ such that:

$$A\mathbf{v} = \lambda\mathbf{v},$$

where λ is the scalar associated with the eigenvector $\mathbf{v}$.

- This equation implies that when matrix A acts on the vector $\mathbf{v}$, it only stretches or shrinks the vector by the factor λ , without changing its direction.

Proof:

Step 2: Rearrange the eigenvalue equation.

- Rearranging the equation $A\mathbf{v} = \lambda\mathbf{v}$, we get:

$$A\mathbf{v} - \lambda\mathbf{v} = 0,$$

or equivalently:

$$(A - \lambda I)\mathbf{v} = 0,$$

where I is the identity matrix of size $n \times n$.

- This equation says that $\mathbf{v}$ is a non-zero vector in the null space of $(A - \lambda I)$.

Step 3: Condition for a non-zero solution.

- For $\mathbf{v} \neq 0$ to be a solution to the equation $(A - \lambda I)\mathbf{v} = 0$, the matrix $(A - \lambda I)$ must be *singular*.
- A matrix is singular if and only if its determinant is zero. Therefore, for the equation $(A - \lambda I)\mathbf{v} = 0$ to have a non-trivial (non-zero) solution, we require:

$$\det(A - \lambda I) = 0.$$

Step 4:

- Conversely, if λ is a root of $\det(A - tI) = 0$, then $A - \lambda I$ is singular. This implies there exists a non-zero vector $\mathbf{v}$ such that:

$$(A - \lambda I)\mathbf{v} = 0 \Rightarrow A\mathbf{v} = \lambda\mathbf{v}.$$

Hence, λ is an eigenvalue of A .

Example 1: Eigenvalues and Eigenvectors of a 3×3 Matrix

Matrix

Consider

$$A_{3 \times 3} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 1: Characteristic Equation

$$\det(A - tI) = \det \begin{bmatrix} 2-t & 0 & 0 \\ 0 & 1-t & 0 \\ 0 & 0 & 1-t \end{bmatrix} = (2-t)(1-t)^2 = 0.$$

Eigenvalues: $t = 2, 1, 1$.

Example 1

Step 2: Eigen Vector Computation

- For $t = 2$: Solve $(A - 2I)\mathbf{v} = 0$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0 \Rightarrow \mathbf{v} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

- For $t = 1$: Solve $(A - I)\mathbf{v} = 0$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0 \Rightarrow \mathbf{v} = c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Example 2

Matrix

Consider the matrix

$$[B]_{2 \times 2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Step 1: Characteristic Equation

$$\det(B - tI) = \det \begin{bmatrix} -t & -1 \\ 1 & -t \end{bmatrix} = (-t)(-t) - (-1)(1) = t^2 + 1 = 0.$$

Eigenvalues: $t = i, -i$ (complex, no real eigenvalues).

Step 2: Conclusion

Since eigenvalues are not real, this matrix has *no real eigenvectors*.

Orthonormal Vectors

Definition

A set of vectors $\{v_1, v_2, \dots, v_k\}$ in $\mathbb{R}^n$ is called **orthonormal** if:

- **Orthogonal:** $v_i^T v_j = 0$ for all $i \neq j$ (vectors are perpendicular),
- **Normalized:** $\|v_i\| = 1$ for all i (vectors have unit length).

Example

Consider vectors in $\mathbb{R}^2$:

$$v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

- $v_1^T v_2 = 0$ (orthogonal)
- $\|v_1\| = \|v_2\| = 1$ (normalized)

Hence, $\{v_1, v_2\}$ is an orthonormal set.

Note:

Two vectors v_i and v_j are **orthogonal** if the **vectors are perpendicular** i.e. $v_i^T v_j = 0$.

Diagonalization of a Square Matrix

Definition

A square matrix $A_{n \times n}$ with real entries is said to be *diagonalizable* if there exists an invertible matrix $P_{n \times n}$ and a diagonal matrix $D_{n \times n}$ such that:

$$A = PDP^{-1}$$

where:

- $P_{n \times n}$ is the matrix whose columns are the eigenvectors of $A_{n \times n}$,
- $D_{n \times n}$ is a diagonal matrix with the corresponding eigenvalues of $A_{n \times n}$ on the diagonal.

Positive Definite Matrix

Definition

A real symmetric matrix A is called **positive definite** if

$$x^T A x > 0 \quad \text{for all nonzero } x \in \mathbb{R}^n.$$

Theorems on Diagonalization of Square Matrices

- **Theorem 1:** A matrix $A_{n \times n}$ is diagonalizable if and only if it has n linearly independent eigenvectors.
- **Theorem 2:** If $A_{n \times n}$ has n distinct eigenvalues, then it is diagonalizable.
- **Theorem 3:** Every real symmetric matrix $A_{n \times n}$ can be diagonalized by an orthogonal matrix: $A = QDQ^T$, where $QQ^T = I$ and D is diagonal.
- **Theorem 4:** Every positive definite matrix is diagonalizable and all eigenvalues of the matrix are positive.

Diagonalization Theorems

Diagonalization Criterion

Theorem: A matrix $A_{n \times n}$ is diagonalizable if and only if it has n linearly independent eigenvectors.

Proof

Assume A has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ corresponding to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.

- Form the matrix $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$. Since the vectors are linearly independent, P is invertible.
- Let D be the diagonal matrix with eigenvalues on the diagonal: $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.
- Then by definition of eigenvectors: $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ for all $i = 1, \dots, n$.
- Writing this for all vectors together:

$$AP = A[\mathbf{v}_1 \ \dots \ \mathbf{v}_n] = [A\mathbf{v}_1 \ \dots \ A\mathbf{v}_n] = [\lambda_1\mathbf{v}_1 \ \dots \ \lambda_n\mathbf{v}_n] = PD.$$

- Multiply both sides by P^{-1} we get $A = PDP^{-1}$.

Proof:

Suppose A is diagonalizable, i.e.,

$$A = PDP^{-1}$$

for some invertible P and diagonal D . Let columns of P be $\mathbf{v}_1, \dots, \mathbf{v}_n$. Then:

$$A\mathbf{v}_i = PDP^{-1}\mathbf{v}_i = PDe_i = P(\lambda_i \mathbf{e}_i) = \lambda_i \mathbf{v}_i,$$

where λ_i is the i -th diagonal entry of D . Thus, $\mathbf{v}_1, \dots, \mathbf{v}_n$ are n linearly independent eigenvectors of A .

Distinct Eigenvalues Implies Diagonalizability

Theorem

If $A_{n \times n}$ has n distinct eigenvalues, then it is diagonalizable.

Proof

Let the n distinct eigenvalues of A be $\lambda_1, \lambda_2, \dots, \lambda_n$.

- *Step 1: Eigenvectors corresponding to distinct eigenvalues are linearly independent.*

Suppose $\mathbf{v}_1, \dots, \mathbf{v}_n$ are eigenvectors corresponding to $\lambda_1, \dots, \lambda_n$. Consider a linear combination:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}.$$

Apply A to both sides:

$$A(c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n) = c_1 A\mathbf{v}_1 + \dots + c_n A\mathbf{v}_n = c_1 \lambda_1 \mathbf{v}_1 + \dots + c_n \lambda_n \mathbf{v}_n = \mathbf{0}.$$

- *Step 2: Construct linear equations for coefficients.*

Subtract λ_1 times the original equation:

$$c_1 \lambda_1 \mathbf{v}_1 + \dots + c_n \lambda_n \mathbf{v}_n - \lambda_1 (c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n) = c_2 (\lambda_2 - \lambda_1) \mathbf{v}_2 + \dots + c_n (\lambda_n - \lambda_1) \mathbf{v}_n = \mathbf{0}.$$

Proof Continued

Step 3: Use induction to show coefficients are zero

Since $\lambda_i \neq \lambda_1$ for $i = 2, \dots, n$, the above implies $c_2 = \dots = c_n = 0$. Similarly, $c_1 = 0$. Hence, the eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly independent.

Step 4: Conclude diagonalizability

By the diagonalization criterion, a matrix with n linearly independent eigenvectors is diagonalizable. Thus, A is diagonalizable.

Powers of a Diagonalizable Matrix

Theorem

If $A_{n \times n}$ is diagonalizable as $A = PDP^{-1}$, then for any integer $k \geq 1$:

$$A^k = PD^kP^{-1}.$$

Sketch of Proof

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1},$$

and by induction:

$$A^k = PD^kP^{-1} \quad \text{for all } k \geq 1.$$

Spectral Theorem

Symmetric Matrix Diagonalization

Every real symmetric matrix $A_{n \times n}$ can be diagonalized by an orthogonal matrix: $A = QDQ^T$, where $QQ^T = I$ and D is diagonal.

Sketch of the Proof of Spectral Theorem

Step 1: Symmetric Matrices have Real Eigenvalues

Let $Av = \lambda v$ with $v \neq 0$.

$$v^T Av = \lambda v^T v.$$

Using symmetry,

$$v^T Av = (Av)^T v,$$

which is real. Thus all eigenvalues of A are **real**.

Step 2: Orthogonality of Distinct Eigenvectors

Let $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$.

$$v_1^T Av_2 = \lambda_2 v_1^T v_2 = (Av_1)^T v_2 = \lambda_1 v_1^T v_2.$$

Thus $(\lambda_1 - \lambda_2) v_1^T v_2 = 0$, so $v_1^T v_2 = 0$. Distinct eigenvalues give **orthogonal** eigenvectors.

Sketch of the Proof of Spectral Theorem

Step 3: Orthonormal Eigenbasis

Each eigenspace has a basis of eigenvectors. Applying Gram–Schmidt gives an **orthonormal** basis. Collect them into

$$Q = [q_1 \ q_2 \ \cdots \ q_n], \quad Q^T Q = I.$$

Step 4: Diagonalization

Since $Aq_i = \lambda_i q_i$,

$$AQ = Q\Lambda,$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Multiply by Q^T :

$$Q^T AQ = \Lambda,$$

so

$$A = Q\Lambda Q^T.$$

Thus A is **orthogonally diagonalizable**.

Gram–Schmidt Theorem

Theorem

Let $v_1, v_2, \dots, v_n$ be linearly independent vectors in $\mathbb{R}^n$. Then there exists an orthonormal set $u_1, u_2, \dots, u_n$ such that

$$\text{span}(u_1, \dots, u_k) = \text{span}(v_1, \dots, v_k) \quad \text{for each } k = 1, 2, \dots, n.$$

Proof

We construct the orthonormal set inductively.

Step 1: Initialize. Set

$$u_1 = \frac{v_1}{\|v_1\|}.$$

Then u_1 is normalized and

$$\text{span}(u_1) = \text{span}(v_1).$$

Step 2: Inductive Step. Suppose $u_1, \dots, u_{k-1}$ have been constructed such that they are orthonormal and

$$\text{span}(u_1, \dots, u_{k-1}) = \text{span}(v_1, \dots, v_{k-1}).$$

Define

$$w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, u_j \rangle u_j.$$

Proof

By construction, w_k is orthogonal to each u_j for $1 \leq j \leq k-1$:

$$\langle w_k, u_j \rangle = \langle v_k, u_j \rangle - \sum_{i=1}^{k-1} \langle v_k, u_i \rangle \langle u_i, u_j \rangle = \langle v_k, u_j \rangle - \langle v_k, u_j \rangle = 0.$$

Since $v_1, \dots, v_n$ are linearly independent, $w_k \neq 0$. Normalize w_k :

$$u_k = \frac{w_k}{\|w_k\|}.$$

Then u_k is a unit vector orthogonal to $u_1, \dots, u_{k-1}$.

Step 3: Span Preservation.

$$\text{span}(u_1, \dots, u_k) = \text{span}(u_1, \dots, u_{k-1}, u_k) = \text{span}(v_1, \dots, v_k),$$

because u_k is a linear combination of $v_1, \dots, v_k$ and vice versa.

Step 4: Completion. Repeat this process for $k = 2, \dots, n$ to obtain an orthonormal set

$$u_1, u_2, \dots, u_n$$

satisfying the theorem.

Example: Diagonalization of a Covariance Matrix

Suppose that the covariance matrix of the random vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix},$$

is given by:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Because covariance matrices are symmetric and positive definite, they can be diagonalized by an orthogonal change of basis.

Eigenvalues of Σ

The upper-left 2×2 block

$$\begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$$

has characteristic polynomial:

$$\lambda^2 - 6\lambda + 1.$$

Thus two eigenvalues are

$$\lambda_1 = 3 + 2\sqrt{2}, \quad \lambda_2 = 3 - 2\sqrt{2}.$$

The last diagonal entry of Σ gives a third eigenvalue:

$$\lambda_3 = 2.$$

So

$$D = \begin{bmatrix} 3 + 2\sqrt{2} & 0 & 0 \\ 0 & 3 - 2\sqrt{2} & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Eigenvectors of Σ

Corresponding eigenvectors are:

$$v_1 = c_1 \begin{bmatrix} 1 \\ -(1 + \sqrt{2}) \\ 0 \end{bmatrix}, \quad v_2 = c_2 \begin{bmatrix} 1 \\ -(1 - \sqrt{2}) \\ 0 \end{bmatrix}, \quad v_3 = c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Assembling them as columns gives the matrix

$$P = \begin{bmatrix} 1 & 1 & 0 \\ -(1 + \sqrt{2}) & -(1 - \sqrt{2}) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

With the eigenvector matrix P and diagonal eigenvalue matrix D , we have

$$\Sigma = PDP^{-1}.$$

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 09: Population Principal Component Analysis

Lecture 03: Mathematical Formulation of Principal Component Analysis



Population Setup and Notation

Random Vector X

Let $X = (X_1, X_2, \dots, X_p)^T$ be a random vector representing the population, where:

- p is the number of variables in the population.

The population mean vector is:

$$\mu = \mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_p])^T$$

and the population covariance matrix is:

$$\Sigma = \text{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^T].$$

Interpretation

Population PCA analyzes the true underlying variability of the population without relying on finite samples.

Goal of Population PCA

Goal

Population PCA seeks a set of uncorrelated linear combinations of the original variables, called principal components, that capture the maximum variance of the population.

Population PCA: Setup

Random Vector

Let

$$X = (X_1, X_2, \dots, X_p)^T$$

be a population random vector with mean μ and covariance matrix Σ .

Definition of Components

Population principal components are defined as

$$Y_i = q_i^T (X - \mu), \quad i = 1, \dots, p.$$

Goal

Find orthogonal directions q_i that:

- maximize the variance of each Y_i
- ensure the components are uncorrelated

Properties of Population PCA

Variance

$$\text{Var}(Y_i) = q_i^T \Sigma q_i.$$

Uncorrelatedness

$$\text{Cov}(Y_i, Y_j) = q_i^T \Sigma q_j = 0 \quad \text{for } i \neq j.$$

Interpretation

The vectors q_i rotate the coordinate system so that the components

$$Y_i = q_i^T (X - \mu)$$

are uncorrelated and ordered by decreasing variance.

First Principal Component

Optimization Problem

$$q_1 = \arg \max_{q \in \mathbb{R}^p} q^T \Sigma q \quad \text{subject to } q^T q = 1.$$

Component

$$Y_1 = q_1^T (X - \mu).$$

Meaning

This direction captures the largest possible variance among all unit-length linear combinations of $X - \mu$.

Second Principal Component

Optimization Problem

$$q_2 = \arg \max_{q \in \mathbb{R}^p} q^T \Sigma q \quad \text{subject to } q^T q = 1, \quad q^T q_1 = 0.$$

Component

$$Y_2 = q_2^T (X - \mu).$$

Uncorrelatedness Constraint

The condition $q^T q_1 = 0$ ensures

$$\text{Cov}(Y_1, Y_2) = 0.$$

General k -th Principal Component

General Optimization Problem

For $k = 1, \dots, p$:

$$q_k = \arg \max_{q \in \mathbb{R}^p} q^T \Sigma q$$

subject to

$$q^T q = 1, \quad q^T q_j = 0 \text{ for all } j < k.$$

Component

$$Y_k = q_k^T (X - \mu).$$

Interpretation

Each new Y_k maximizes variance while being uncorrelated with all earlier components.

Interpretation of Population PCA

Geometric View

PCA constructs a new coordinate system aligned with directions of maximal variability.

Transformation

The vectors $q_1, \dots, q_p$ form an orthogonal rotation, producing components

$$Y_i = q_i^T (X - \mu).$$

Key Insight

The components are uncorrelated and arranged in decreasing order of variance, each revealing new, nonredundant structure in the distribution of X .

Theorem 1: Population Principal Components are Uncorrelated and Have Variances Equal to Eigenvalues

Theorem

Let $X = (X_1, X_2, \dots, X_p)^T$ be a population random vector with mean μ and covariance matrix Σ . Let (λ_i, q_i) be the eigenvalue-eigenvector pairs of Σ , ordered such that $\lambda_1 \geq \dots \geq \lambda_p \geq 0$. Then, the population principal components defined by

$$Y_i = q_i^T (X - \mu), \quad i = 1, \dots, p$$

satisfy:

- $\text{Var}(Y_i) = \lambda_i$
- $\text{Cov}(Y_i, Y_j) = 0$ for $i \neq j$

Proof of Theorem 1

Proof

- The population covariance matrix is:

$$\Sigma = \text{Cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^T]$$

- Eigenvalue decomposition:

$$\Sigma = Q\Lambda Q^T$$

where $Q = [q_1, \dots, q_p]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$.

- **Theorem (Quadratic Form Maximization):** Let B be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$ and eigenvectors $e_1, \dots, e_p$. Then

$$\max_{\|q\|=1} q^T B q = \lambda_1,$$

achieved **only** when $q = e_1$.

- **Applying the Theorem on $B = \Sigma$ we get:**

$$q_1 = e_1, \quad Y_1 = e_1^T (X - \mu), \quad \text{Var}(Y_1) = \lambda_1.$$

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Proof Continued

- After choosing $q_1 = e_1, \dots, q_{i-1} = e_{i-1}$, maximize variance over unit vectors orthogonal to previous components:

$$\max_{\substack{\|q\|=1 \\ q \perp q_1, \dots, q_{i-1}}} q^\top \Sigma q.$$

- Theorem (Quadratic Form Maximization under Orthogonality Constraints):** For symmetric B with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$ and eigenvectors $e_1, \dots, e_p$, for $k = 1, \dots, p-1$:

$$\max_{\substack{\|q\|=1 \\ q \perp e_1, \dots, e_k}} q^\top B q = \lambda_{k+1},$$

achieved **only** when $q = e_{k+1}$.

- Applying the Theorem on $B = \Sigma$ and $k = i-1$:**

$$q_i = e_i, \quad Y_i = e_i^\top (X - \mu), \quad \text{Var}(Y_i) = \lambda_i.$$

Proof of Theorem 1

Proof Continued:

- Variance of the i -th component:

$$\text{Var}(Y_i) = \text{Var}(q_i^T (X - \mu)) = q_i^T \Sigma q_i = \lambda_i$$

- Covariance between different components:

$$\text{Cov}(Y_i, Y_j) = q_i^T \Sigma q_j = q_i^T (\lambda_j q_j) = \lambda_j q_i^T q_j = 0, \quad i \neq j$$

since eigenvectors are orthonormal.

Population Principal Components

Population Principal Components

The i -th principal component of the population is:

$$Y_i = q_i^T (X - \mu), \quad i = 1, \dots, p,$$

where q_i is the eigenvector of Σ corresponding to eigenvalue λ_i .

Theorem 2: Proportion of Total Population Variance Explained by the k -th Component

Theorem

Let $Y_1, \dots, Y_p$ be the population principal components of X with eigenvalues $\lambda_1, \dots, \lambda_p$. Then, the total population variance is

$$\text{Var}_{\text{total}} = \text{tr}(\Sigma) = \sum_{i=1}^p \lambda_i$$

and the proportion of total variance explained by the k -th component is

$$\frac{\lambda_k}{\lambda_1 + \dots + \lambda_p}.$$

Proof of Theorem 2

Proof

- Total population variance is sum of variances of original variables:

$$\text{Var}_{\text{total}} = \sum_{i=1}^p \text{Var}(X_i) = \text{tr}(\Sigma) = \lambda_1 + \cdots + \lambda_p$$

- Variance explained by the k -th principal component is λ_k
- Therefore, the proportion of total variance explained by the k -th component is

$$\frac{\lambda_k}{\lambda_1 + \cdots + \lambda_p}$$

Theorem 3: Correlation Between Population PCs and Original Variables

Theorem

Let $Y_i = q_i^T(X - \mu)$ be the i -th population principal component, and X_k be the k -th original variable with variance σ_{kk} . Then, the correlation is:

$$\rho_{Y_i, X_k} = \frac{q_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i, k = 1, \dots, p$$

where q_{ik} is the k -th element of eigenvector q_i corresponding to λ_i .

Proof of Theorem 3

Proof

- By definition, correlation:

$$\rho_{Y_i, X_k} = \frac{\text{Cov}(Y_i, X_k)}{\sqrt{\text{Var}(Y_i)\text{Var}(X_k)}}$$

- Covariance:

$$\text{Cov}(Y_i, X_k) = \text{Cov}(q_i^T(X - \mu), X_k) = q_i^T \Sigma e_k = \lambda_i q_{ik}$$

where e_k is the k -th standard basis vector.

- Variances:

$$\text{Var}(Y_i) = \lambda_i, \quad \text{Var}(X_k) = \sigma_{kk}$$

- Substituting:

$$\rho_{Y_i, X_k} = \frac{\lambda_i q_{ik}}{\sqrt{\lambda_i \sigma_{kk}}} = \frac{q_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}$$

Variance and Covariance of Population Principal Components

Variance of Components

$$\text{Var}(Y_i) = q_i^T \Sigma q_i = \lambda_i, \quad i = 1, \dots, p$$

Covariance Between Components

$$\text{Cov}(Y_i, Y_j) = q_i^T \Sigma q_j = 0 \quad \text{for } i \neq j$$

Interpretation

Each principal component captures variance along an orthogonal direction in the population, ensuring uncorrelated, independent directions of maximum variability.

Eigenvalue Problem for Population PCA

Eigenvalue Equation

The principal components are obtained by solving:

$$\Sigma q_i = \lambda_i q_i, \quad i = 1, \dots, p$$

- λ_i is the variance explained by the i -th principal component.
- q_i is the corresponding population eigenvector (principal direction).

Ordering

The components are ordered by decreasing eigenvalues:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$$

so that Y_1 captures the most population variance, Y_2 the second-most, and so on.

Step-by-Step Population PCA Procedure

- 1 Define the population mean $\mu = \mathbb{E}[X]$ and population covariance $\Sigma = \text{Cov}(X)$.
- 2 Solve the eigenvalue problem $\Sigma q_i = \lambda_i q_i$ for eigenvectors q_i and eigenvalues λ_i .
- 3 Order eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$ and select the corresponding eigenvectors.
- 4 The population principal components are $Y_i = q_i^T (X - \mu)$.

First and Second Population Principal Components

First Principal Component

$$Y_1 = q_1^T (X - \mu)$$

captures the largest amount of population variance λ_1 .

Second Principal Component

$$Y_2 = q_2^T (X - \mu)$$

is orthogonal to Y_1 and captures the second-largest population variance λ_2 .

Generalization

Subsequent components Y_i are orthogonal to all previous ones, capturing the remaining variance.

Variance Explained by Population Principal Components

Proportion of Variance

The proportion of total population variance explained by the k -th component is:

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}$$

where λ_k is the eigenvalue corresponding to Y_k .

Total Variance

$$\text{Total Population Variance} = \text{tr}(\Sigma) = \sum_{i=1}^p \lambda_i$$

Interpretation

This helps decide how many components suffice to capture most population variability.

Correlation Between Population PCs and Original Variables

Population Correlation

The correlation between the i -th population PC Y_i and the original variable X_k is:

$$\rho_{Y_i, X_k} = \frac{q_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i, k = 1, \dots, p$$

where σ_{kk} is the variance of X_k and q_{ik} is the k -th entry of eigenvector q_i .

Interpretation

This shows how each population principal component is related to the original variables.

Population PCA Example

Population Covariance Matrix

Suppose the population covariance of three random variables X_1, X_2, X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues and Eigenvectors

$$\lambda_1 = 5.83, \quad \lambda_2 = 2, \quad \lambda_3 = 0.17$$

$$q_1 = \begin{bmatrix} 0.383 \\ -0.924 \\ 0 \end{bmatrix}, \quad q_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad q_3 = \begin{bmatrix} 0.924 \\ 0.383 \\ 0 \end{bmatrix}$$

Population PCs

$$Y_1 = 0.383(X_1 - \mu_1) - 0.924(X_2 - \mu_2), \quad Y_2 = X_3 - \mu_3, \quad Y_3 = 0.924(X_1 - \mu_1) + 0.383(X_2 - \mu_2)$$

Proportion of Total Population Variance Explained

Total Variance

$$\text{tr}(\Sigma) = 1 + 5 + 2 = 8$$

$$\sum_{i=1}^3 \lambda_i = 5.83 + 2 + 0.17 = 8$$

Proportion Explained

$$Y_1 : \frac{5.83}{8} \approx 0.73, \quad Y_1 + Y_2 : \frac{5.83 + 2}{8} \approx 0.98$$

Conclusion

The first two population principal components capture 98% of the population variance.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
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Module 09: Population Principal Component Analysis

Lecture 04: Mathematical Foundations of Principal Component Analysis



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$$\text{Var}(Y_i) = q_i^T \Sigma q_i.$$

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$$\text{Cov}(Y_i, Y_j) = q_i^T \Sigma q_j = 0 \quad \text{for } i \neq j.$$

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are uncorrelated and ordered by decreasing variance.

First Principal Component

Optimization Problem

$$q_1 = \arg \max_{q \in \mathbb{R}^p} q^T \Sigma q \quad \text{subject to } q^T q = 1.$$

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This direction captures the largest possible variance among all unit-length linear combinations of $X - \mu$.

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Each new Y_k maximizes variance while being uncorrelated with all earlier components.

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PCA constructs a new coordinate system aligned with directions of maximal variability.

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The vectors $q_1, \dots, q_p$ form an orthogonal rotation, producing components

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The components are uncorrelated and arranged in decreasing order of variance, each revealing new, nonredundant structure in the distribution of X .

Theorem 1: Population Principal Components are Uncorrelated and Have Variances Equal to Eigenvalues

Theorem

Let $X = (X_1, X_2, \dots, X_p)^T$ be a population random vector with mean μ and covariance matrix Σ . Let (λ_i, q_i) be the eigenvalue-eigenvector pairs of Σ , ordered such that $\lambda_1 \geq \dots \geq \lambda_p \geq 0$. Then, the population principal components defined by

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satisfy:

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Proof

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achieved **only** when $q = e_1$.

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Proof Continued

- After choosing $q_1 = e_1, \dots, q_{i-1} = e_{i-1}$, maximize variance over unit vectors orthogonal to previous components:

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- Variance of the i -th component:

$$\text{Var}(Y_i) = \text{Var}(q_i^T (X - \mu)) = q_i^T \Sigma q_i = \lambda_i$$

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since eigenvectors are orthonormal.

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Proof

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Proof

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$$\rho_{Y_i, X_k} = \frac{\text{Cov}(Y_i, X_k)}{\sqrt{\text{Var}(Y_i)\text{Var}(X_k)}}$$

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The correlation between the i -th population PC Y_i and the original variable X_k is:

$$\rho_{Y_i, X_k} = \frac{q_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i, k = 1, \dots, p$$

where σ_{kk} is the variance of X_k and q_{ik} is the k -th entry of eigenvector q_i .

Interpretation

This shows how each population principal component is related to the original variables.

Population PCA Example

Population Covariance Matrix

Suppose the population covariance of three random variables X_1, X_2, X_3 is:

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues and Eigenvectors

$$\lambda_1 = 5.83, \quad \lambda_2 = 2, \quad \lambda_3 = 0.17$$

$$q_1 = \begin{bmatrix} 0.383 \\ -0.924 \\ 0 \end{bmatrix}, \quad q_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad q_3 = \begin{bmatrix} 0.924 \\ 0.383 \\ 0 \end{bmatrix}$$

Population PCs

$$Y_1 = 0.383(X_1 - \mu_1) - 0.924(X_2 - \mu_2), \quad Y_2 = X_3 - \mu_3, \quad Y_3 = 0.924(X_1 - \mu_1) + 0.383(X_2 - \mu_2)$$

Proportion of Total Population Variance Explained

Total Variance

$$\text{tr}(\Sigma) = 1 + 5 + 2 = 8$$

$$\sum_{i=1}^3 \lambda_i = 5.83 + 2 + 0.17 = 8$$

Proportion Explained

$$Y_1 : \frac{5.83}{8} \approx 0.73, \quad Y_1 + Y_2 : \frac{5.83 + 2}{8} \approx 0.98$$

Conclusion

The first two population principal components capture 98% of the population variance.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 09: Population Principal Component Analysis

Lecture 05: Population Principal Component Analysis for Normal Distribution



Principal Component Analysis for Multivariate Normal Distribution

Setup

Consider a random vector

$$X \sim N_p(\mu, \Sigma),$$

a multivariate normal with mean vector μ and covariance matrix Σ .

Density on Ellipsoids

The density of X is constant on ellipsoids defined by:

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = c^2.$$

These ellipsoids are centered at μ and aligned with axes given by the eigenvectors e_i of Σ .

Eigenvalues and Ellipsoid Axes

Eigenpairs Define Ellipsoid Axes

The eigenpairs (λ_i, e_i) , $i = 1, \dots, p$, define the ellipsoid axes:

- e_i : directions,
- λ_i : scaling factors (eigenvalues).

Ellipsoid Equation

Points on the ellipsoid satisfy:

$$c^2 = \sum_{i=1}^p \frac{(e_i^\top (x - \mu))^2}{\lambda_i}.$$

Principal Components

The coordinate system aligned with eigenvectors e_i defines the principal components.

Principal Components as Coordinates

Definition

For a point x , principal components are defined as:

$$y_i = e_i^\top (x - \mu).$$

Interpretation

These coordinates express x relative to the ellipsoid axes.

Simplified Ellipsoid Equation

The ellipsoid condition becomes:

$$c^2 = \sum_{i=1}^p \frac{y_i^2}{\lambda_i}.$$

Geometric Interpretation for Bivariate Normal

Setup for $p = 2$

For two variables, principal components y_1, y_2 correspond to rotated axes of the data ellipse.

Ellipse and Density

The ellipse equation

$$(x - \mu)^\top \Sigma^{-1} (x - \mu) = c^2$$

defines contours of equal density.

Coordinate Rotation

Principal components rotate the coordinate system so that axes align with the ellipse.

This simplifies the shape of the density contours.

Summary

Key Points

- Principal components diagonalize the covariance matrix Σ .
- They define new coordinate axes aligned with directions of maximum variance.
- Projection onto principal components transforms ellipsoids into spheres in the new space.
- PCA helps uncover independent sources of variability, simplifying data structure analysis.

Principal Component Analysis for Multivariate Standard Normal Distribution

Motivation

Before performing PCA, we may first standardize the variables so that:

- each feature has mean 0,
- each feature has variance 1,
- PCA is based on **correlation** instead of covariance.

Standardized Variables

For a p -dimensional vector $X = (X_1, \dots, X_p)^T$,

$$Z_i = \frac{X_i - \mu_i}{\sqrt{\sigma_{ii}}}, \quad i = 1, \dots, p,$$

where σ_{ii} is the variance of X_i .

Matrix Notation for Standardization

Diagonal Standard Deviation Matrix

Let

$$V^{1/2} = \begin{pmatrix} \sqrt{\sigma_{11}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\sigma_{pp}} \end{pmatrix}.$$

Then the standardized vector Z is

$$Z = (V^{1/2})^{-1}(X - \mu).$$

Covariance of Standardized Data

$$\text{Cov}(Z) = (V^{1/2})^{-1}\Sigma(V^{1/2})^{-1} = \rho,$$

where ρ is the **correlation matrix** of X .

PCA of Standardized Variables

Key Idea

PCA of Z is equivalent to performing eigen-decomposition of the correlation matrix ρ :

$$\rho e_i = \lambda_i e_i.$$

Here:

- λ_i are eigenvalues,
- e_i are eigenvectors,
- eigenvalues measure variance of principal components.

Principal Components

The i -th principal component of standardized variables is:

$$Y_i = e_i^T Z = e_i^T (V^{1/2})^{-1} (X - \mu).$$

Variance of Principal Components

Variance of Each Standardized Variable

$$\text{Var}(Z_1) + \cdots + \text{Var}(Z_p) = p.$$

Variance of Each PC

Eigenvalues give the PC variances:

$$\text{Var}(Y_i) = \lambda_i.$$

Total Standardized Variance

$$\sum_{i=1}^p \lambda_i = p.$$

Correlation Between PCs and Standardized Variables

Correlation Formula

The correlation between Y_i and standardized variable Z_k is

$$\rho_{Y_i, Z_k} = e_{ik} \sqrt{\lambda_i},$$

where e_{ik} is the k -th entry of eigenvector e_i .

Interpretation

- Large $|e_{ik}|$ means variable Z_k contributes strongly to PC i .
- Sign shows direction (positive/negative loading).

Proportion of Variance Explained

Standardized Case

Total standardized variance is p , so the proportion explained by the k -th PC is:

$$\frac{\lambda_k}{p}$$

Meaning

This tells us how much of the total standardized variation is captured by each principal component.

Summary

- Standardizing transforms X into Z with mean 0 and variance 1.
- PCA on Z uses the **correlation matrix** ρ , not Σ .
- Eigenvalues of ρ give PC variances.
- Eigenvectors give linear combinations of standardized variables.
- Total variance is p , so proportion explained is λ_i/p .

Example using Bivariate Normal Distribution

Data Simulation

We simulate data from a bivariate normal distribution with two variables X_1 and X_2 , where the correlation between them is high 0.9 (close to 1). This is represented by the covariance matrix:

$$\Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$

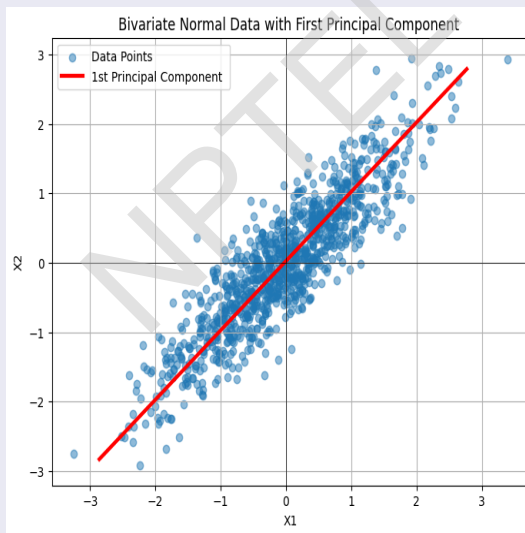
The data is generated from a multivariate normal distribution with mean $\mu = [0, 0]$ and covariance matrix Σ . A total of 1000 samples are simulated:

$$X \sim \mathcal{N}(\mu, \Sigma), \quad n = 1000.$$

Visualization

Visualizing the Data and First Principal Component

The plot below shows the bivariate data along with the first principal component (major axis of the ellipse) indicated by the red line segment:



Principal Component Analysis (PCA)

PCA and Eigenvalues

- PCA is used to decompose the variance in the data along the *principal components* (PCs).
- The first principal component (PC1) corresponds to the direction of maximum variance.
- The second principal component (PC2) corresponds to the direction of minimum variance.
- PCA computes the *eigenvalues* and *eigenvectors* of the covariance matrix.
- Eigenvalues: λ_1, λ_2 (where $\lambda_1 \geq \lambda_2$).
- The sum of the eigenvalues gives us the *total variance* in the dataset.

Variance Based on Eigenvalues

Total Variance (Eigenvalues)

The total variance in the dataset is the sum of the eigenvalues from PCA:

$$\text{Total Variance} = \lambda_1 + \lambda_2$$

Since the data is highly correlated, most of the variance will be captured by the first principal component.

Eigenvalue Interpretation

- The first eigenvalue (λ_1) corresponds to the variance along the first principal component (major axis of the ellipse).
- The second eigenvalue (λ_2) corresponds to the variance along the second principal component (minor axis of the ellipse).

Variance Based on Projections

Projection onto the First Principal Component

- The second type of variance is based on projecting each data point onto the first principal component (major axis of the ellipse).
- This projection gives us a scalar value for each point that represents how far it lies along the direction of maximum variance.
- The projection of a point $\mathbf{x}_i$ onto the first principal component $\mathbf{v}_1$ is:

$$p_i = \mathbf{v}_1^T \mathbf{x}_i$$

- The distance of each point from the origin (mean-centered) is:

$$\text{Distance from origin} = |p_i|$$

Variance of Projected Distances

- The variance of these distances gives us the spread of the data along the major axis of the ellipse:

$$\text{Variance of Projected Distances} = \text{Var}(p_i)$$

Comparison of Variances

Expected Differences in Variance

- *Eigenvalue-based variance*: The sum of the eigenvalues captures the total variance in the 2D data space, accounting for both the major and minor axes of the ellipse.
- *Projection-based variance*: The variance based on the first principal component is generally smaller than the total variance, as it only captures variance along the major axis (first principal component).
- Therefore, we do not expect the variances to be the same. The eigenvalue-based variance captures both axes, while the projection-based variance captures only the variance along the first principal component.

Expected Outcome

- The projection-based variance will be smaller because:

$$\lambda_1 > \lambda_2$$

PCA Summary Table

Principal Component	Eigenvalue	Explained Variance Ratio
PC1	1.756895	0.944614
PC2	0.103014	0.055386

Sum of Eigenvalues (Total Variance): 1.8599

Variance of Projected Distances (Along PC1): 0.6685

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
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